Simple interaction rules qualitatively reproduce Calhoun's Universe 25: a pattern-oriented agent-based model of population collapse

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Abstract

The aim of this work is to show that a few simple interaction rules qualitatively reproduce the population collapse observed by Calhoun in Universe 25, a closed mouse colony with unlimited resources that stopped breeding and died out. We propose an agent-based model in which crowding lowers an individual reproductive competence that juveniles acquire from their neighbours, and we compare its output with seventeen patterns reported by Calhoun. When adults cannot recover from crowding damage, the model reproduces the collapse: growth that stops without resource limits, natality that ceases near the peak, cohorts that never breed, and extinction. When adults recover above a threshold rate, the same rules yield a colony that persists, as observed in other confined mouse populations. The collapse requires neither spatial structure nor the social rules, which add specific patterns such as a persistent class of withdrawn non-breeders.

Keywords: agent-based model, pattern-oriented modelling, density dependence, social learning, reproductive suppression, population collapse, Universe 25

1. Introduction

Can simple, local interaction rules reproduce what Calhoun (1973) observed in Universe 25? Four pairs of house mice, in an enclosure with food, water and nest space in excess and with no predators or emigration, grew

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to about 2 200 animals, well below the 3 840 that the nest boxes could shelter. Growth then ceased, the last young to survive were born shortly after the peak, and the colony declined without recovery. Along the way Calhoun described a breakdown of social behaviour: mothers that abandoned their young, withdrawn males pooled in the centre of the floor, and the unwounded, non-breeding "beautiful ones". Mice of a companion colony, moved to uncrowded enclosures late in its decline, bred very poorly even with healthy partners (Marsden, 1972). Calhoun read the collapse as a failure of social roles rather than of resources. The experiment had a single replicate, its extinction was projected, and its popular reading, "density causes pathology", departed from Calhoun's own emphasis on the quality of social interaction (Ramsden and Adams, 2009). Other confined colonies of house mice with food in excess also stopped growing below the limit set by food (Southwick, 1955a; Crowcroft and Rowe, 1957), in Southwick's colonies mainly because litters failed to survive (Southwick, 1955b). In deermice, density-dependent suppression of breeding is reversible: animals removed from asymptotic populations breed again (Terman, 1973). Intrinsic regulation through social behaviour is an old theme of small-mammal ecology (Christian, 1950; Chitty, 1967).

Individual-based models are the natural tool to ask how such population-level outcomes emerge from the behaviour of individuals (Railsback and Grimm, 2019). A single sample path of a strongly stochastic process cannot be fitted, but it can be read as a set of patterns. Pattern-oriented modelling uses several patterns, observed at different levels of organisation, as filters for the structure of a model (Grimm et al., 2005). We translate Calhoun's report into seventeen falsifiable patterns, score each of them in every stochastic run, and compare alternative model structures: null models, the removal of each rule, a model in which adults recover, and well-mixed versions without space. Unlike a recent quantitative model that switches on deviations when the total population exceeds critical values (Gwizdałła and Duś, 2026), and unlike accounts based on game theory (Mantzaris, 2025) or on fitted unimodal curves (Roman, 2025), no rule here depends on population size. To our knowledge, the closest individual-based models of rodent populations address cycles driven by sociality, dispersal and predation (Radchuk et al., 2016), not a collapse with unlimited resources.

The answer is that a few local processes suffice; by simple interaction rules we mean all the rules of the model. Crowding damage that gates breeding and care, and a long life span, suffice for the collapse of Universe 25 when

the non-breeding state of adults is nearly irreversible (Fig. 1); the collapse occurs without space and without the four social rules of the model, in a core model and in a well-mixed colony in which a short-lived control persists. The social rules add specific patterns: a juvenile window with social learning makes the late-born cohorts fail, small retreat sites lodge a persistent class of withdrawn non-breeders, and crowd aversion sets the timing of the end of natality. Space adds aggregation that leaves nest sites empty at a peak far below capacity, and a prolonged stagnation phase. When adults recover on their own above a threshold rate (two recovery parameters changed; Section 2.1), the same rules give a colony whose breeding resumes after the peak and which survives, with large, persistent, slowly damped oscillations instead of collapsing. Adult irreversibility, the assumption with the least empirical support, and a long life span are thus the conditions that separate the two outcomes, and the recovery that the collapse tolerates decreases with colony size. Earlier confined colonies offer a few held-out patterns, which we use as a consistency check of the reversible branch, not as a validation.

Here we ask which patterns of Universe 25 simple rules reproduce, with which ingredients, under which conditions they give persistence instead of collapse, and which patterns they do not reproduce. Every simulated number used here is backed by archived runs in the code repository, and those on the conditions of the mechanism are summarised in the Supplementary Material (SM, Section S7).

2. Methods

2.1. Model

The model description follows the ODD protocol (Grimm et al., 2020); the full ODD and the code are in the Supplementary Material (SM, Section S2). The model is individual based and stochastic, with one time step per week. Individuals live in a closed arena of $|G| = L \times L$ cells ($L = 30$). A cell stands for a nest site or local group: it is the unit at which crowding is experienced and mates are met. Each individual has an age a, a sex, a position and a *reproductive competence* $s \in [0, 1]$, a phenomenological scalar that summarises the capacity to breed, the capacity to rear young and the quality of the individual as a demonstrator for juveniles. There are no resources and no predators, and no individual knows the population size. The simple interaction rules of the title are all the rules below: the core processes, among them crowding damage and senescence, which suffice for the collapse

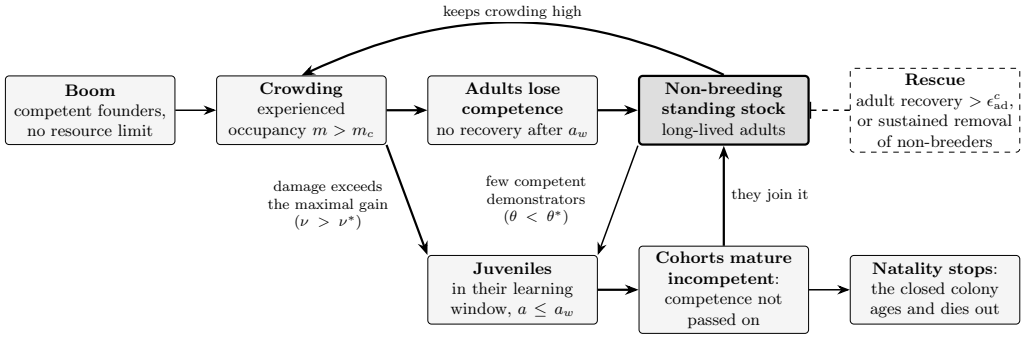


Figure 1: Causal scheme of the collapse. Thick arrows form the reinforcing loop. Crowding makes adults lose their reproductive competence. After the sensitive window ($a > a_w$) the loss is permanent, and the long-lived non-breeders (a standing stock of non-breeding adults) keep crowding high: in space during the aggregated growth, in the well-mixed limit by holding the density above ν^* (sufficiency conjectural; Section 3.6). Juveniles in their window are then damaged faster than they can learn. They also learn from few competent demonstrators ($\theta < \theta^*$, a precursor), so they mature incompetent and join the standing stock, and natality stops. Whether adults recover is the condition of the mechanism (dashed): above a threshold ϵ_{ad}^c of adult recovery the colony persists (Fig. 3); a sustained removal of non-breeders also rescues it. Protecting only the juveniles, with perfect demonstrators or a crowding-free nursery, does not (798 of 800 colonies went extinct; SM, Section S7). Every step acts on individuals.

(Section 3.3), and four social rules, which add specific patterns. Table 1 lists the key parameters.

Core processes. Crowding damage. Each week, after moving, competence is updated with the occupancy m_i of the cell of i, itself included:

$$s_i \leftarrow \min\{1, \max[0,\ s_i + g_i - \gamma\,(m_i - m_c)_+]\}, \tag{1}$$

where g_i is the weekly gain (below), m_c the tolerated group size and γ the damage per extra occupant. In Universe 25 social disruption arose while food and water were in excess (Calhoun, 1973), so the rule is interference without a resource. It responds to the crowding that an individual experiences, not to the density $N/|G|$. *Competence gates breeding and care.* A reproductive female that shares her cell with a reproductive male (chosen at random) conceives with probability $p_0 s_f s_m$; each pup survives birth with probability $q_0 s_f$ and starts with a fraction w of its mother's competence,

$$p_{\text{conc}} = p_0\, s_f\, s_m, \qquad p_{\text{surv}} = q_0\, s_f, \qquad s_{\text{pup}} = w\, s_f. \tag{2}$$

The last relation is a maternal effect (Ginzburg and Taneyhill, 1994), not inheritance; it is an assumption of C1, not an ingredient of the mechanism ($w = 0$ with faster learning gives the same outcome, Section 3.4). The rule encodes the failures that Calhoun reported in the crowded phase: males that ceased courting, and mothers that abandoned their litters. *Senescence and attraction.* Death is senescent, with weekly hazard $[1 + e^{-k(a-a_{\max})}]^{-1}$, independent of s; females breed between $a_{\min}$ and a_{rep} weeks of age. These values follow Universe 25: menopause at about 560 days, many mice alive at about 800 days, and withdrawn animals that were "physically healthy" (Calhoun, 1973). Each week every individual moves to a cell y of its 3×3 neighbourhood (own cell included) with probability proportional to

$$W_i(y) = \big[1 + \alpha\, \chi_i(y)\big]\; e^{-\kappa(1-s_i)^p n_y}\; \big[1 + \pi(1 - s_i)^q\, \mathbf{1}_{\mathcal{R}}(y)\big], \tag{3}$$

where n_y is the number of other individuals in y and $\chi_i(y)$ the summed affinity of i for them; affinities grow by proximity and decay with a memory of about 100 weeks. The first factor, attraction to acquaintances, aggregates the colony, in the spirit of Calhoun's "behavioural sink" (Calhoun, 1962).

Four social rules. The four social rules are labelled as in the code and in the figures. *Sensitive window (R1).* Competence can be gained only up to age a_w, a window that spans weaning and maturity ($g_i = 0$ for $a_i > a_w$), while crowding damage acts at every age. Mice removed from a declining companion colony bred very poorly in uncrowded enclosures: after two months without reproduction, half of the females conceived with normal males and 5 of about 100 pups survived (Marsden, 1972). The rule is the standard form of a developmental sensitive window (Fawcett and Frankenhuis, 2015). Of all the assumptions, adult irreversibility has the least empirical support and the largest effect; a variant multiplies the adult gain by a factor $\epsilon_{\text{ad}} \in [0, 1]$, the adult recovery factor (C1: $\epsilon_{\text{ad}} = 0$). *Social learning of competence (R2).* Inside the window the gain is set by the neighbours,

$$g_i = r\big[(1 - \lambda) + \lambda\, \bar{s}_{V(i)}\big] \qquad (a_i \le a_w), \tag{4}$$

where $\bar{s}_{V(i)}$ is the mean competence of the other individuals of the 3×3 block around i (zero if there are none); C1 has $\lambda = 1$, purely social learning. A pup starts at $ws_f \le 0.2$ and can gain at most $ra_w = 1.2$, but only among competent demonstrators. The rule follows Calhoun's observation that "by midway in phase C essentially all young were prematurely rejected by their

Table 1: Key parameters of the reference model C1 (SM, Table S3, for all parameters and their justification). The short-lived control differs in $a_{\max} = 60$, $a_{\text{rep}} = 50$, $r = 0.01$, $\lambda = 0$, $a_w = \infty$, $w = 1$, $\alpha = 1.4$ and has neither crowd aversion nor retreat sites.

Symbol	Value	Meaning	Symbol	Value	Meaning
L	30	side of the arena ($\|G\| = 900$ cells)	r	0.1	maximal weekly gain of competence
N_0	200+200	founders, age 4 weeks, $s = 1$	a_w	12 weeks	sensitive window (R1)
m_c	8	tolerated occupancy	λ	1	weight of social learning (R2)
γ	0.1	damage per extra occupant	ϵ_{ad}	0	adult recovery factor
p_0	0.3	weekly conception probability	w	0.2	maternal transmission
q_0	0.95	pup survival at $s_f = 1$	α	4	attraction to acquaintances
litter	3–10, mean 6.55	litter size	κ, p	0.15, 4	crowd aversion (AV)
$a_{\min}$, a_{rep}	6, 80 weeks	reproductive ages of females	f_R, c_R	0.3, 2	retreat sites (R7)
$a_{\max}$, k	120 weeks, 0.15	senescence	π, q	100, 4	preference for retreat sites (R7)

mothers"; it is consistent with the non-genomic transmission of maternal care in rats (Francis et al., 1999) and the social learning of pup retrieval in mice (Carcea et al., 2021), which support the social transmission of maternal care, not of male sexual competence, which the model extends by hypothesis.

Crowd aversion (AV) and retreat sites (R7). The second factor of Eq. (3) makes deteriorated animals ($s \lesssim 0.3$) avoid crowded cells, and the third makes them prefer retreat sites, a fraction f_R of the cells that hold at most c_R individuals. They follow Calhoun's descriptions of withdrawn males that pooled in the centre, "beautiful ones" that never fought, and withdrawn females that retreated to the high, less-preferred nest boxes (Calhoun, 1973).

Order of processes and demography. Each week, in this order: every individual moves (Eq. 3, simultaneously, with the capacity of the retreat sites enforced); affinities are updated; occupancies are counted; competence is updated (Eqs. 1 and 4); females mate in their cell, conceive and give birth to the pups that survive, which are placed in the mother's cell; individuals age and die; observables are recorded. The founders, 200 females and 200 males of age 4 weeks with $s = 1$, are placed at random. Four simplifications of mouse demography have measured effects (SM, Fig. S2). There is no gestation or interbirth interval: conception and birth occur in the same weekly step, with

a ceiling of about two pups per female and week; with a refractory period of three weeks after each birth C1 still collapses (60 of 60 runs extinct with either of two seed sets; joint outcome 59 of 60 with an independent seed and 60 of 60 with the production seeds), and the extinction time changes by +1.0 [−2.6, 4.6] weeks (paired with the production seeds). Pups move and add to crowding from birth; keeping them in the maternal cell for three weeks lowers the joint outcome (below) to 0.83. The founders form a synchronous cohort, and 69% of all births are theirs; initial ages spread over 4–80 weeks give 0.63. There is no mortality before senescence; a background mortality of 0.003 per week lowers the joint outcome to 0.55 and 0.008 to 0.06, in both cases through the timing of the end of natality, while up to 0.05 per week does not rescue the colony (SM, Section S7). Longevity is part of the mechanism, not a detail (Section 3.6).

Model structures compared. The reference model C1 has the core processes and the four social rules. The *core model* keeps the core processes but replaces the four social rules by neutral values: no window, individual recovery at a slow rate ($r = 0.01$) at every age, full maternal transmission ($w = 1$), no crowd aversion and no retreat sites. A null model has no crowding damage ($\gamma = 0$). The *short-lived control* (Table 1) regenerates competence individually; in space 89 of its 200 colonies are still alive at week 1 000, cycling with a period of 162 [155, 170] weeks. The *reversible-suppression null* is C1 with the two recovery parameters changed: adults recover at a fraction $\epsilon_{\mathrm{ad}} = 0.8$ of the juvenile rate, and $\lambda = 0.9$ instead of 1 keeps social learning but lets them recover partly on their own, at the asocial rate $\epsilon_{\mathrm{ad}}(1 - \lambda) = 0.08$ of the juvenile rate. With $\lambda = 1$, $\epsilon_{\mathrm{ad}} = 0.8$ lies inside the transition (10 of 20 colonies die out at $L = 30$); the null lies above the rescue threshold at every size simulated (Section 3.2). The parameters that set the form of the rules were chosen to reproduce the most qualitative patterns with the fewest rules; they were not fitted to Calhoun's curves. In the *well-mixed* variants every individual moves each week to a uniformly random cell, so that attraction, aversion and the preference for retreat sites have no effect and the retreat sites act only as a capacity; cell occupancies are Poisson with mean $N/|G|$. A female then mates either within her cell, as in space (*local mating*), or with a reproductive male drawn from the whole arena (*global mating*), which removes any difficulty of finding a mate.

2.2. Patterns, evaluation and statistical design

From Calhoun's report we extracted seventeen qualitative, falsifiable patterns and four anti-patterns, each translated into an operational criterion on simulated time series (SM, Table S1; codes in parentheses). Growth: latency, exponential growth and deceleration (O1–O3, scored only with a separate founder protocol from four pairs), and a socially functional growth phase (O10b). Peak: a peak far below capacity with empty nest sites (O4), adults that deteriorate before it (O10), aggregation with free space (O14), and coexisting crowded and withdrawn non-breeders (O13), with a persistent withdrawn class (O13p) born late (O13b). Decline: late-born cohorts that fail to mature (O11) and rarely conceive (O11b), natality that ceases near the peak (O5), no recovery of natality (O6), extinction (O7), a long senescent decline (O8) to an old, female-biased remnant (O9), total infant mortality (O15), and the failure of transplanted late animals to breed (O12). We call O5, O6, O7 and O10 the *collapse envelope*. Calhoun's dates enter only through dimensionless ratios of the time and size of the peak. A criterion returns PASS, MARGINAL (one step from its threshold), FAIL or not applicable for each run, and an ensemble is summarised by the fraction of passing runs with a 95% Wilson score interval. The joint outcome O17p requires every applicable essential pattern, no anti-pattern and a persistent withdrawn class. Every run was scored with the evaluator preregistered before the production runs (a time-stamped plan in the code repository, 1 October 2026; SM, Section S3) and with a primary evaluator, fixed before those runs, that dates the peak at the start of the plateau rather than at $\arg\max N$; the two are reported side by side (SM, Table S4). One option, the end of natality in O5 at the week by which 99% of the births have occurred, was chosen after exploratory runs had shown that the last sustained birth fails O5 in about half of the C1 runs (with that reference O17p is 0.485, 97 of 200 production runs). Square brackets give 95% intervals unless stated otherwise.

The unit of analysis is the independent stochastic replicate. The design comprises production ensembles of the model structures ($n = 200$ each), a factorial ablation of the rules, transplant experiments (60 replicates per arm), a sweep of the adult recovery factor at three arena sizes, robustness variants, lattice sizes, and a global sensitivity analysis (Morris screening, two Latin hypercubes and Sobol' indices); seven confirmatory hypotheses were tested with Holm correction, and extinction times are summarised by Kaplan–Meier estimates (SM, Sections S4 and S5).

Use of AI in the research. Starting from the author's preliminary implementation and draft rules, Claude (Anthropic) agents, used through Claude Code (v2.1; Claude Opus 5.5 and Claude Fable 5.1 for the research tasks, Claude Sonnet 5.5 for literature searches and auxiliary programming), reimplemented and extended the code, ran the simulations and the statistical analyses, and searched the literature. The author set the objectives, the qualitative targets and the interaction rules, decided every change of scope and approved each stage. The code has an automated test suite; scripts in the code repository regenerate every figure and number from the archived outputs; independent Claude agents reviewed the work, checking the code against the text and reproducing key results; and the history is kept under version control.

3. Results

3.1. The reference colony follows Calhoun's sequence

The time course of C1 has the four phases of Universe 25 in Calhoun's order (Fig. 2). On a time axis relative to the peak of each system (t_{pk}: the start of the plateau in the model, median week 34; day 560 in Universe 25), births start at $0.09\,t_{\mathrm{pk}}$ and growth slows at $0.31\,t_{\mathrm{pk}}$ (Universe 25: 0.19 and 0.56). The mean competence of the adults falls below 0.5 at week 13, while the colony is still growing. Births stop near the peak: 99% of them have occurred by week 40 and the last one comes at week 59 [46, 76] (median and 10–90% range over 200 runs). The shapes differ after the peak. The model stays on a plateau for about $1.5\,t_{\mathrm{pk}}$ and then falls along a sigmoid as the closed cohort ages, whereas Universe 25 declined from its peak; the model dies out at $4.97\,t_{\mathrm{pk}}$, week 169 (Kaplan–Meier median; 95% interval [167, 171]; 10–90% range [157, 186]), against a projected $3.18\,t_{\mathrm{pk}}$ for Universe 25. The peak uses 0.36 of the damage-threshold capacity $K_\gamma = m_c|G|$ and leaves 16% of the cells empty (Universe 25: about 20% of the nest boxes empty, an agreement in sign, since $|G|$ and m_c were not calibrated). The model starts from 400 young adults rather than four pairs, so the comparison is of shape, not of timing (doubling times: SM, Section S1).

3.2. Adult recovery decides the outcome

Whether adults made non-breeding by crowding recover decides between collapse and persistence (Fig. 3). If adults regain competence at a fraction ϵ_{ad} of the juvenile rate, every colony still dies out for $\epsilon_{\mathrm{ad}} \leq 0.02$ (2 880 of

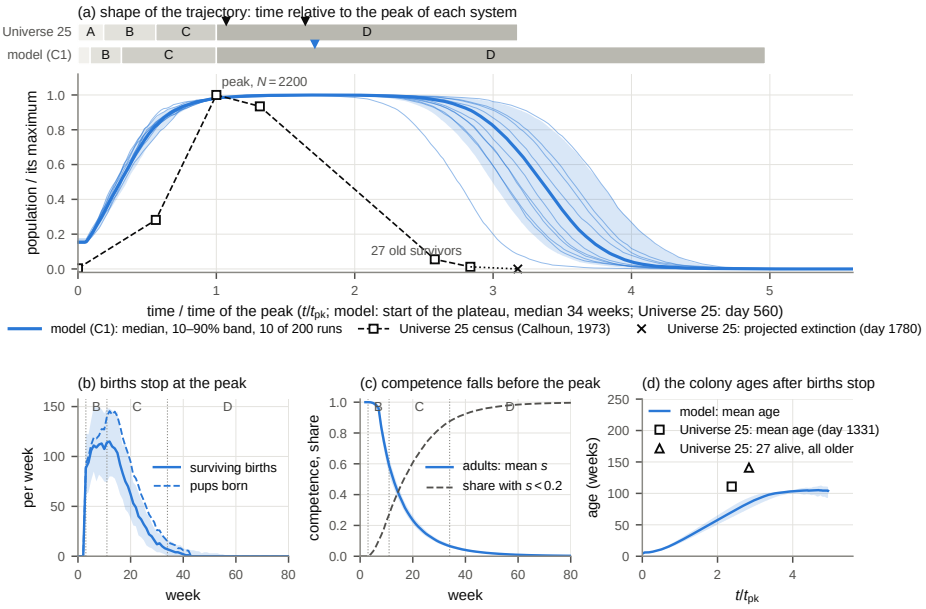


Figure 2: The reference colony and Universe 25. (a) Population relative to its maximum against time relative to the peak of each system, t/t_{pk} (t_{pk}: start of the plateau, $N \geq 0.98 \max N$, in each run of the model, median week 34; day 560 in Universe 25). Model: C1, median, 10–90% band and 10 of 200 colonies, rescaled run by run. Squares: census of Universe 25 (Calhoun, 1973); cross: Calhoun's projected extinction (day 1 780). Ribbons: phases A–D (A, adjustment; B, growth; C, slower growth; D, death phase) as dated by Calhoun and, for the model, as measured in each run (medians): first births, halving of the growth rate of $\ln N$, start of the plateau and extinction. Triangles: last surviving young and last conception in Universe 25, last birth in the model. No curve is fitted. (b) Pups born and pups surviving birth per week. (c) Mean reproductive competence of the adults and share of all animals with $s < 0.2$. (d) Mean age of the living animals against t/t_{pk}; symbols: mean age of the survivors of Universe 25 on day 1 331, and the 27 survivors on day 1 588, all older than 987 days. Bands: 10–90% of colonies. Dotted lines in (b, c): phase boundaries.

2 880 colonies at $L = 15$ and 30, in space and well mixed, for every λ; SM, Table S7), so the collapse is not an artefact of the edge $\epsilon_{\mathrm{ad}} = 0$. Above a threshold $\epsilon_{\mathrm{ad}}^{c}$ the colony persists. In space the threshold falls with the side of the arena without visible saturation: from 0.133 [0.113, 0.157] to 0.085 [0.073, 0.099] and 0.050 [0.037, 0.063] between $L = 15$, 30 and 60 when recovery is individual ($\lambda = 0$), and from 0.97 [0.88, >1] to 0.73 [0.67, 0.81] and 0.53 [0.46, 0.62] when it is purely social ($\lambda = 1$; profile-likelihood intervals of the midpoint; SM, Table S8). Three sizes do not determine its

limit for $L \to \infty$, which could be zero. Well mixed with global mating it also falls (0.81, 0.51, 0.30, again with non-overlapping intervals); with mating in the cell it is resolved only to the grid of ϵ_{ad} (0.80, 0.6–0.8, 0.60; overlapping intervals), compatible with the deterministic limit, 0.585. With $\lambda \geq 0.9$ no recovery rescues the well-mixed colony (all but one of 4 860 colonies die out), because an adult whose recovery is social finds no competent demonstrators. The robust statement is therefore the edge: the collapse requires that adults recover asocially at most slowly, $\epsilon_{\mathrm{ad}}(1-\lambda) \approx 0$, at the scale of the experiment.

The reversible-suppression null, C1 with $\epsilon_{\mathrm{ad}} = 0.8$ and $\lambda = 0.9$ (asocial recovery at 0.08 of the juvenile rate), persists in all 200 colonies to week 1 000. It keeps every spatial and deterioration pattern of C1 (200 of 200 runs each; Fig. 5, first 600 weeks), and its natality nearly ceases after the peak (to 0.1% of its maximum, median), so the two outcomes share the first half of the sequence; but breeding then resumes. The collapse envelope separates them: the end of natality, the absence of recovery and the extinction fail in all 200 runs. Nor does the colony settle. After a plateau from week 34 to about week 100 (median) it falls to 0.15 of the peak (week 141) and then cycles with a period of 148 [147, 149] weeks (peak to peak; mean over runs, 143–152 across runs, 10–90%). The oscillations are persistent and slowly damped. The peak-to-trough amplitude decays from 0.85 of the first peak to 0.49, 0.42 and 0.39 in the next three cycles and then only slowly (-0.013 [-0.017, -0.008] of the first peak per 100 weeks over cycles 3–5, about 3% of the amplitude), so that at week 1 000 the colony still oscillates between about 0.40 and 0.80 of the first peak. Whether the amplitude levels off (a geometric fit gives 0.35 [0.33, 0.37]) or keeps decaying cannot be decided within 1 000 weeks. The autocorrelation of N over weeks 300–1000, corrected for the shrinking overlap of the lagged series, is 0.90 at one, two and three periods (0.71, 0.51 and 0.34 uncorrected), which rules out a fast-relaxing quasi-cycle. In all 200 runs the evaluation thus finds cycles with recovery of births and no logistic saturation: the trajectory differs in form, not only in level. With initial ages spread over 4–80 weeks the null falls to 0.23 [0.21, 0.25] of the peak and cycles with the same period, so the cycle is not an artefact of the synchronous founding cohort.

Earlier confined colonies, not used to build or tune the model, allow only a consistency check of this branch, since its persistence follows from the choice of ϵ_{ad} above the threshold. With food in excess those colonies stopped growing below the limit set by food, mainly through the failure of litters to survive (Southwick, 1955a,b); confined populations of several rodent

species regulated their density (Lidicker, 1965); and confined deermice of asymptotic populations did not breed until removed from them (Terman, 1973). Scored on the null (re-run with the archived seeds to record every litter), litter loss is concentrated on the plateau: the fraction of litters with no surviving pup rises from 0.05 during growth to 0.24 [0.24, 0.25] (0.09 in C1) and falls back to 0.07–0.10 afterwards. But the fall of natality is mainly a fall of conception (73% of its logarithmic decline), so this pattern is reproduced only in part. Most females born on the plateau never breed (0.89 do not conceive by 48 weeks of age), but so do those of C1 (0.96). In C1 no female is born after the plateau, whereas in the null those born during the decline breed (only 0.37 do not conceive by 48 weeks; 0.72 of those born from week 300 on); what separates the outcomes is whether adults resume breeding (definitions and sources of these held-out patterns: SM, Section S7). The quick recovery of deermice removed from the asymptotic population is reproduced by construction: phase-D animals of C1 moved to an enclosure where adults recover individually found a colony in 51 of 60 trials (Fig. 6e), whereas Calhoun's transplants bred very poorly (Marsden, 1972), the pattern of the irreversible branch.

3.3. The collapse needs neither space nor the four social rules

Removing space requires a control (Fig. 4e). When every animal is relocated at random each week but mates only within its cell, the deformation is extinctive in itself: the short-lived control, which in space is still alive in 89 of 200 colonies at week 1 000, dies out in 200 of 200 well-mixed colonies (median 120 weeks, earlier than C1, 147), through a mate-finding Allee effect after the overshoot. The test therefore rests on global mating, in which a female mates with a reproductive male drawn from the whole arena. There the control lives far longer than C1 and persists in larger arenas (extinct by week 1 000: 81, 1 and 0 of 200 at $L = 30$, 45 and 60; at $L = 30$ all colonies die out by week 3 000), whereas C1 dies out in all 600 colonies at 138–140 weeks, almost independently of L. The core model without the four social rules collapses as well, in space and in both well-mixed variants (all 1 000 colonies; median extinction at 136–166 weeks). The core model and the short-lived control share the same slow individual recovery at every age ($r = 0.01$); well mixed, where attraction (α, 4 in the core and 1.4 in the control) plays no role, only the long life span of the core makes it collapse. The cessation of breeding and the extinction thus need neither space nor the four social rules: they need crowding damage and long-lived adults that recover slowly, if at all

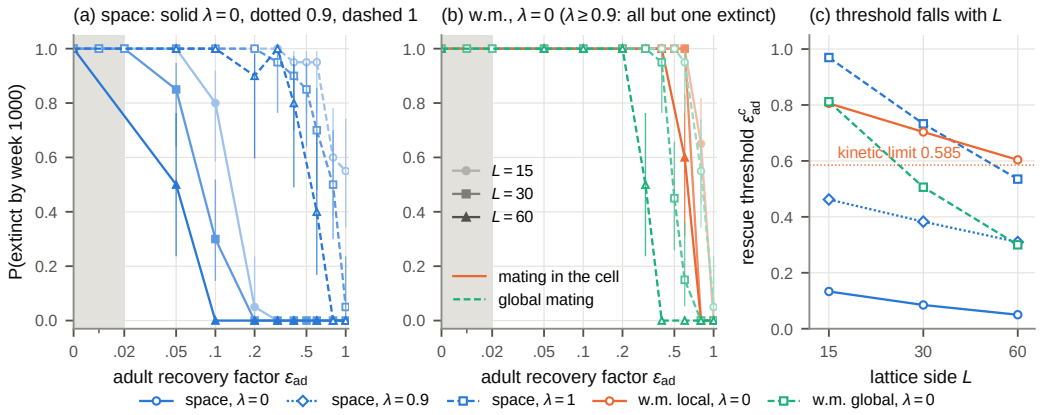


Figure 3: Adult recovery decides the outcome. Probability that a C1 colony is extinct by week 1 000 against the adult recovery factor ϵ_{ad} (fraction of the juvenile rate). (a) Spatial model (with retreat sites), individual ($\lambda = 0$, solid), mostly social ($\lambda = 0.9$, dotted) or purely social recovery ($\lambda = 1$, dashed). (b) Well-mixed model, $\lambda = 0$, mating in the cell (orange) or global (green); with $\lambda \geq 0.9$ all but one of the 4 860 well-mixed colonies (three mating rules) die out. Shades and markers: $L = 15$, 30, 60 (20, 20 and 10 colonies per point; 95% Wilson intervals where $0 < P < 1$). Grey band: every colony collapses. (c) Rescue threshold ϵ^c_{ad} (midpoint of a logistic fit) against L; dotted: value of the kinetic equation, 0.585. Data: SM, Tables S7 and S8.

(Section 3.2). A component Allee effect remains with global mating, because a female paired with an incompetent male conceives with probability $p_0 s_f s_m$.

Without space the end of natality and the extinction are the same, but the sequence is not. With mating in the cell, growth stays fast and competent until the peak (a socially functional growth phase in 200 of 200 runs, against 4 of 200, and 194 marginal, in space), the peak is 2.3 times denser (6.6 against 2.9 animals per cell), and competence then collapses at once, so that phase C lasts two weeks (weeks 24–26), against 23 weeks in space and about 35 in Universe 25. With global mating phase C lasts 11 weeks (weeks 7–18), so the compression comes from local mating, not from the absence of space. Space also changes how the colony is organised. At the spatial peak the mean occupancy is 2.9 animals per cell, but it is aggregated (variance-to-mean ratio 2.53 [2.51, 2.55]; maximum about 16) and 18% [18, 19] of the animals sit in cells above m_c (SM, Figs. S3 and S4): the colony crowds into pools next to empty cells and lodges a class of withdrawn non-breeders, isolated adults with $s < 0.2$ that appear already during growth and sit mostly in the retreat sites; well mixed the occupancy is homogeneous and almost nobody stays

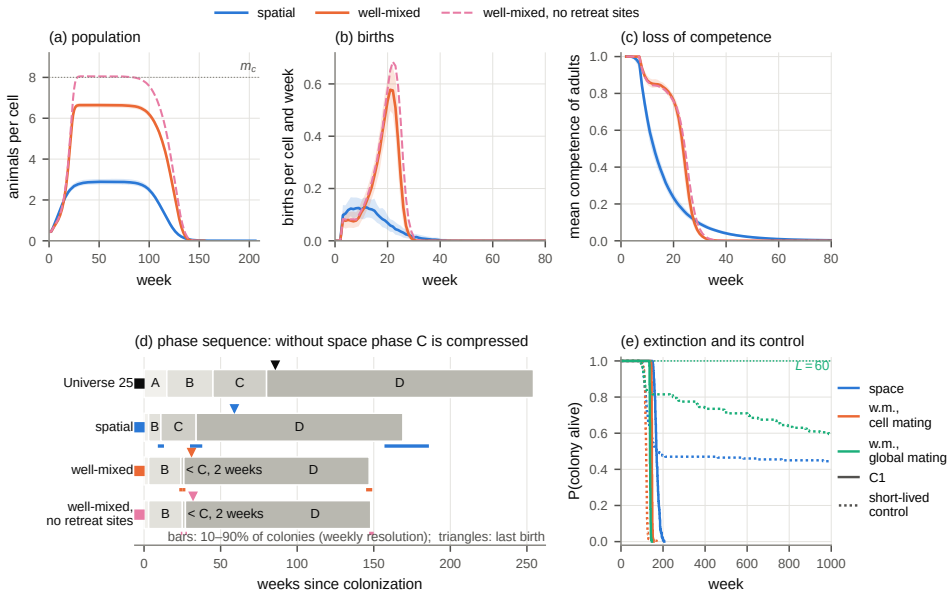


Figure 4: Extinction with and without space, and its control. Spatial C1 (blue, $n = 200$) and well-mixed C1, in which every animal is relocated at random every week and mates in its cell (orange, $n = 200$; retreat sites act only as a capacity; pink dashed, without retreat sites, $n = 200$); $L = 30$, 400 initial animals. (a) Animals per cell; dotted line: the tolerated crowding m_c. (b) Births per cell and week. (c) Mean reproductive competence of the adults. Medians with 10–90% bands. (d) Phase sequence of Universe 25 and of each model, dated as in Fig. 2 (medians; bars below: 10–90% of the colonies at weekly resolution; triangles: last birth). With mating in the cell phase C lasts two weeks. (e) Kaplan–Meier survival to week 1 000 of C1 (solid) and of the short-lived control (dotted) in space, well mixed with mating in the cell, and well mixed with global mating ($L = 30$; the control with global mating also at $L = 60$, where no colony dies out). Global mating appears only in panel (e). Maps of one colony at its peak and in its decline: SM, Fig. S4; full sequences: Supplementary Videos S1–S3.

isolated. That class cannot form under random relocation, so its absence there is a consistency check, not a finding.

3.4. Which rule produces which pattern

The pattern matrix filters structures through the collapse envelope (Fig. 5). The envelope passes in every column with crowding damage and long-lived adults that do not recover fast, except that adult deterioration before the peak is only partial in the well-mixed core (0.78 of runs with mating in the cell and 0.18 with global mating): with global mating the core

reproduces the end of natality and the extinction, but not the deterioration of adults before the peak. The envelope fails in the two reversible structures, the reversible-suppression null and the short-lived control. Without crowding damage the colony explodes (every run reaches the numerical cap of 15 000 animals at weeks 26–29, median 27). The spatial patterns, a peak far below capacity with empty nest sites and aggregation with free space, pass in every spatial column with crowding damage and in no well-mixed one. All essential patterns jointly hold in 0.995 [0.97, 1.00] of the runs of C1 (199 of 200) and in none of the other columns. The senescent nature of the decline holds by construction (there is no social mortality), but its duration criterion passes in 0.62 of the C1 runs; the persistence of the withdrawn class follows from the capacity of the retreat sites, which equals the isolation threshold; the failure of transplanted adults follows from R1; and in the well-mixed columns the coexistence of crowded and withdrawn non-breeders is arithmetic, not behaviour (Fig. 5).

The ablations name the rule behind each pattern (SM, Fig. S1). Without the sensitive window (R1, adults recovering socially) the envelope is lost (extinction in 0.23 of the runs); without social learning (R2) only the failure of the late cohorts; without retreat sites (R7) only the persistence of the withdrawn class; without crowd aversion only the timing clause of the end of natality (0.82) and the duration of the decline. The core model keeps the envelope in space and loses the cohort failure, the coexistence of classes and their persistence. Over two Latin hypercubes of 15 factors, 353 of 400 parameter points (0.88, 95% interval 0.85–0.91) reproduce the collapse envelope, 168 add the failure of the late cohorts, and 9 reproduce every essential pattern (0.022). The collapse is generic once adults do not recover; what is rare is its conjunction with the spatial and class patterns. Maternal transmission is not needed ($w = 0$ with faster learning gives 1.00), and at fixed seeding density the joint outcome rises from 0.82 at $L = 15$ to 1.00 at $L \geq 45$ (SM, Fig. S2).

Of the seven confirmatory predictions fixed before the production runs, three fail with each evaluator (SM, Section S4). Q2, that each rule is necessary, fails because without crowd aversion the joint outcome stays at 0.79 [0.71, 0.86] (0.90 with the preregistered evaluator) against the predicted ≤ 0.5; Q5, a band of coexisting crowded and withdrawn non-breeders centred at the preferred occupancy, is refuted in the opposite direction; and Q6, that the window weighs less than the learning rate, fails with the primary evaluator and holds with the preregistered one. Q1, the failure of transplanted

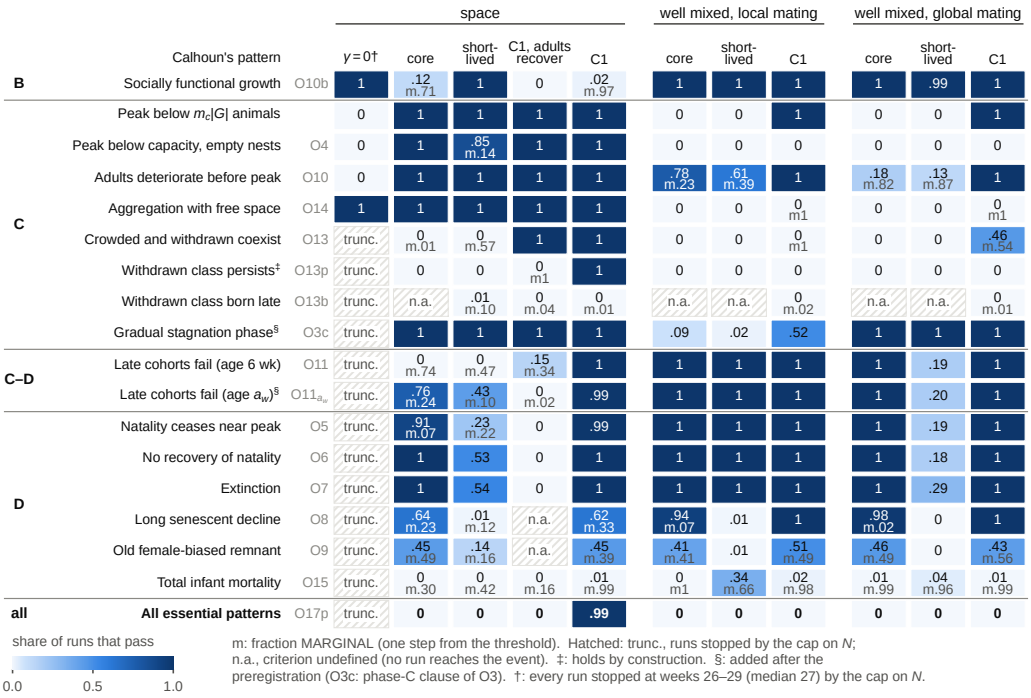


Figure 5: Which model structures reproduce Calhoun's patterns. Colour and upper number: fraction of runs that pass the operational criterion (primary evaluator, $n = 200$ per column, $L = 30$, $T = 600$; Wilson half-width ≤ 0.07); lower number (m): fraction MARGINAL, one step from the threshold. Columns: in space, no crowding damage ($\gamma = 0$; †every run stops at weeks 26–29, median 27, when N reaches the numerical cap of 15 000, so only criteria decided before the cap are shown), the core model (crowding damage, maternal transmission, senescence and attraction, without R1, R2, AV and R7), the short-lived control, C1 with adults that recover ($\epsilon_{\mathrm{ad}} = 0.8$, $\lambda = 0.9$) and C1; well mixed with mating in the cell, and with global mating, for the core model, the control and C1. Each column comes from one ensemble and seed set (SM, Table S6). The first row is empirical (peak below $m_c|G|$). In the well-mixed columns the coexistence of crowded and withdrawn non-breeders is fixed by the capacity of the retreat sites, and the joint outcome is 0 by construction (a withdrawn class cannot persist under random relocation). ‡: holds by construction. §: added after the preregistration (the phase-C clause of O3, $t_D - t_{BC} \geq 0.1\, t_D$; O11 at age a_w).

late animals, holds with the primary evaluator but not with the preregistered one, which takes the transplants at 1.75 times the week of $\arg\max N$, later in the plateau: 12 of 60 groups found a colony ($P_T = 0.20$ [0.12, 0.32]). Q3, Q4 and Q7 hold with both.

3.5. Behavioural patterns

The individual-level patterns of Calhoun's report appear in the model, each with its ingredient (Fig. 6). In space, crowded and withdrawn non-breeders coexist and the withdrawn class persists: at week 60, 18% of the adults are crowded non-breeders, 28% withdrawn and 17% persistently withdrawn; well mixed, isolation is brief and random (57%, 9% and 0.2%). Maternal care fails with the competence of the mothers: the share of pups that survive birth falls through the growth phase in both variants. The late-born cohorts never breed: 35% of the females of the first spatial cohort ever conceive, 11% of those born in week 13 and 6% of those born in week 20 (in a two-cell universe run in parallel with Universe 25, 18% of 148 females born in the last 50 days of phase C had ever conceived by a median age of 334 days; Calhoun, 1973, p. 85). The transplants of Marsden (1972), whose animals bred very poorly rather than not at all, are reproduced qualitatively: groups taken in the growth phase found a colony in 52 of 60 trials, and those taken in the death phase in none, alone or with healthy partners (0/60 and 0/120), by the irreversibility built into R1 (Section 3.2).

3.6. The mechanism

The collapse has the same logic in both structures, a non-breeding state that crowding induces and that adults cannot leave, but the step that drives it differs. In space, aggregation crowds the adults during growth: the crowding experienced by juveniles, $\gamma\,\mathrm{E}(m - m_c)_+$, exceeds the maximal gain r only during the growth phase, when the adults lose their competence within weeks; with $\epsilon_{\mathrm{ad}} = 0$ they never regain it, and the cohorts that follow have neither the uncrowded window nor competent neighbours. In the well-mixed limit, the long-lived non-breeders form a standing stock that keeps the mean crowding above a threshold $\nu^* = 6.92$, at which the expected damage to a juvenile equals r; in the kinetic equation, the limit of the well-mixed model at fixed density, the first cohorts fail through damage inside the window before demonstrators become scarce (SM, Section S7). A second threshold, the proportion $\theta^* = (1 - w)/(ra_w) = 2/3$ of neighbours with $s > 0$, a threshold for the transmission of competence between generations, is only a precursor: in the simulations the mean proportion around juveniles crosses it before natality falls below 5% of its maximum, by a median of 15 weeks in space and 3 well mixed, also in colonies that later persist. Which of these statements are definitions, analytical arguments, conjectures or simulation results is listed in SM Section S7. Once the cohorts mature incompetent, natality stops and

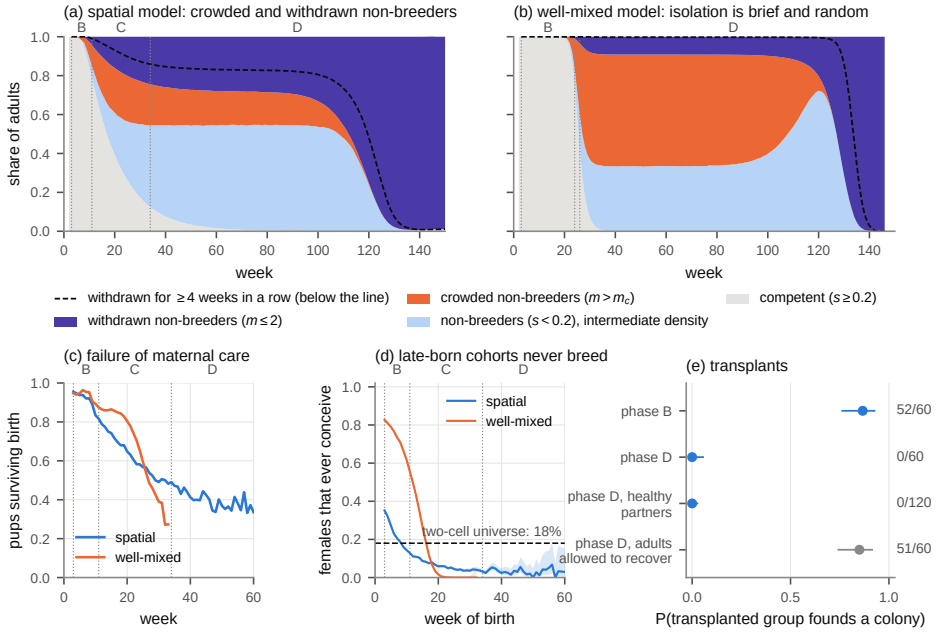


Figure 6: Behavioural patterns. (a, b) Composition of the adults (mean over living colonies) in the spatial and the well-mixed C1: competent ($s \geq 0.2$), and non-breeders ($s < 0.2$) that are crowded ($m > m_c$), withdrawn (at most two animals in the cell) or at intermediate density; below the dashed line, adults withdrawn for at least four consecutive weeks. (c) Share of pups that survive birth, which falls with the competence of the mother (births per conception divided by the mean litter size). (d) Share of the females born in each week that ever conceive, with 95% Wilson bands (well mixed: 83%, 43% and 2% for the cohorts born in weeks 3, 13 and 20); dashed line: 18% of the late phase-C females of Calhoun's two-cell universe had ever conceived, a comparison of order (different colony and definition). (e) Transplants (60 groups of four males and four females per arm, 95% Wilson intervals): a group founds a colony when it reaches 80 animals within 150 weeks in an empty enclosure; phase-D animals are taken at $1.75\,t'_{\mathrm{pk}}$ (primary evaluator). Phase-D animals moved to an enclosure where adults may recover (no R1 and R2, individual recovery at rate 0.1) found a colony in 51 of 60 trials. Dotted lines: phase boundaries of the spatial model (Fig. 2).

the closed colony dies out after the residual life span of the last cohort, 111 weeks after the last birth in space (median) and about 117 well mixed. In the well-mixed limit two independent conditions make the non-breeding state absorbing: no adult recovery ($\epsilon_{\mathrm{ad}} = 0$) or purely social learning ($\lambda = 1$), since an adult whose gain is social cannot recover among non-breeders. Nor is the collapse a mate-finding Allee effect (Gascoigne et al., 2009): well mixed, C1 collapses with mating in the cell, in the 3×3 block or over the whole arena (50 of 50 colonies under each at $L = 15$–60; 95% interval $[0.93, 1]$), whereas the short-lived control persists with global mating for $L \geq 45$ (SM, Table S9).

4. Discussion

4.1. Collapse and persistence from the same rules

Simple local rules, none of which knows the size of the population, qualitatively reproduce much of what Calhoun observed in Universe 25 when adults made non-breeding by crowding do not recover; when adults recover on their own above a threshold rate, the colony resumes breeding and persists, oscillating rather than levelling off. The pattern-oriented comparison of structures assigns the ingredients. The collapse needs crowding damage and long-lived adults that do not recover; it survives every change of structure that we applied, including the removal of the four social rules and of space, except adult recovery and a short life span. The four social rules and space add specific patterns: the failure of the late-born cohorts (social learning in a window), a persistent withdrawn class (retreat sites), the timing of the end of natality (crowd aversion), and empty nest sites at a low peak with a prolonged stagnation (aggregation).

The mechanism connects old ideas of small-mammal ecology. Social stress from crowding depresses reproduction (Christian, 1950), and an individual state carried across a delay makes density dependence delayed (Ginzburg and Taneyhill, 1994). The model adds a developmental window in which competence is learned socially (Fawcett and Frankenhuis, 2015; Carcea et al., 2021) and shows that, with a long life span and a non-breeding state that adults cannot leave, these close a loop: the adults whose competence crowding has removed keep the crowding that prevents the next cohort from acquiring it. The loop acts on a specific class, the long-lived non-breeders, which explains why indiscriminate mortality does not rescue the colony while their sustained removal does (SM, Section S7). The social component of the model is co-presence and the transmission of a state, not social roles: dominance, male

territoriality and reproductive skew, central to Calhoun's account and to the social organisation of confined wild mice (Crowcroft and Rowe, 1963), are not modelled.

4.2. Conditions of the mechanism

Irreversibility. Density-dependent suppression of breeding in rodents is usually considered reversible. In house mice, urine of grouped females delays puberty in young females (Drickamer, 1977), an effect that fades within ten days of ungrouping (Coppola and Vandenbergh, 1985). These signals act on the puberty of juvenile females, the analogue of the juvenile window of the model, which is already reversible; direct evidence on the recovery of adults that have stopped breeding is limited to deermice (Terman, 1973). With adult recovery switched on, the same rules keep the colony alive, oscillating, instead of collapsing it (Section 3.2). Near-irreversibility is supported only by the transplants from Calhoun's colonies (Marsden, 1972), whose animals were several hundred days old, when reproductive senescence of BALB/c females is already relevant; in C1 the failure of the transplants is reproduced by construction, not as independent evidence. The conclusion is therefore conditional: simple rules give the collapse of Universe 25 if adults made non-breeding by crowding do not recover on their own and live long (well mixed with purely social recovery no colony is rescued even at the juvenile rate, because no competent demonstrators are left), and a persisting colony if they do recover. The threshold of adult recovery is the quantity that an experiment on reversibility should estimate.

Scale. Universe 25 was a single colony of one size. At fixed seeding density the model reproduces the patterns more often in larger arenas, but the adult recovery that the collapse tolerates in space decreases with the arena, with no sign of saturation between $L = 15$ and 60 (Fig. 3c). Three sizes do not determine the limit of this threshold for an infinite arena, which may be zero, so for any adult recovery above zero the spatial collapse may be a finite-size effect; only the irreversible case is supported at every size simulated. Extinction itself is certain for any closed colony with crowding damage, because crowding caps the number of animals that can breed and there is no immigration; what distinguishes collapse from persistence is that the extinction time of C1 grows only logarithmically with the size of the colony (SM, Section S7). The mechanism is thus established at the scale of the experiment;

a much larger closed colony would need an even less reversible non-breeding state.

Timing. The conjunction of all patterns is sensitive to when natality ends: with a background mortality of 0.003 per week, or with initial ages spread over 4–80 weeks, it holds in 0.55 and 0.63 of the runs, only through the timing clause of the end-of-natality criterion. The collapse itself is not sensitive: background mortality up to 0.05 per week does not rescue C1 (SM, Section S7), a gestation-like refractory period of three weeks leaves it unchanged (60 of 60 runs extinct; Section 2.1), and with spread initial ages the reversible null still falls to 0.23 of its peak and cycles (Section 3.2).

4.3. Limits of the spatial layer

The spatial patterns are obtained in a regime of mobility opposite to that of Universe 25. The ratio of the mixing time to the deterioration time, a Damköhler number, is about 1.7×10^2 in C1 and at most 4×10^{-3} in Universe 25, where mice crossed the enclosure daily (SM, Section S2): crossing the lattice by diffusion takes about 1.7×10^3 weeks. In Universe 25 the classes arose among mobile animals through behaviour (male territories, possession of nest boxes, aggression); in the model they arise from aggregation under low mobility. The size of the withdrawn class is moreover set by a parameter: from week 30 the 270 retreat sites, with 540 places, are about 94% occupied, so the class (416 adults at the peak in the colony of SM Fig. S4) is bounded by $f_R c_R |G|$. The proportion of withdrawn animals is thus an input of the model, not an output. The collapse, which also occurs well mixed, does not depend on this regime; the classes do.

4.4. Testable predictions

Adult recovery. At the scale of the experiment ($L = 30$), a closed colony collapses if fewer than about 1% of the adults of reproductive age breed again from 30 weeks after natality ends, and persists if more than about 9% do; the tolerated recovery decreases with colony size. It is measurable as the return of oestrous cycling, mating and maternal care in adults removed from high density, as a function of age and exposure. *Sensitive window.* Cohorts reared at high density during the window fail even with competent demonstrators (798 of 800 colonies); cross-fostering or translocation at different ages and densities would locate the window. *Removal of non-breeders.* Periodically removing a quarter to a half of the non-breeding adults rescues the colony,

while a single removal or indiscriminate mortality does not, a prediction for the management of captive breeding colonies of social rodents (SM, Section S7). *Retreat sites.* The withdrawn class scales with the number of small nest sites, $f_R c_R |G|$; manipulating the number of small nest boxes tests it. These predictions concern *Mus* in confinement. Elsewhere they depend on the reversibility of adult suppression, which has been measured directly only in deermice (Terman, 1973).

4.5. What is not reproduced

The founding phase is reproduced only in an exploratory protocol with $a_{\max} = 200$ weeks (about 3.8 years), a life span implausible for *Mus*. The simulated decline is a boom of one or two generations, not Calhoun's multigenerational course: 69% of the births are to the founders, and the net reproductive number falls from 3.8 to 0.40 in one generation. The persistent withdrawn class is made of refugees of the early boom, not of a late-born cohort like the "beautiful ones"; a variant that ties withdrawal to maternal socialisation produces a later-born class at a demographic cost (O17p 0.705). Total infant mortality is absent, the female bias of the remnant holds in fewer than half of the runs (0.45), and aggression is not modelled. The persisting colonies cycle with a period of about 150 weeks instead of levelling off, and the litter-loss pattern of earlier colonies is reproduced only in part. The same Calhoun patterns were used to choose the rules and to report them, so their fulfilment shows that they are possible under local rules, not that they were predicted; only the comparison with earlier confined colonies was held out, and it was defined after the production runs. A quantitative model fitted to the census (Gwizdałła and Duś, 2026) reproduces the founding phase and the timing, which ours does not, through switches on the total population; the two approaches agree that replicate variability is large and that Calhoun's trajectory is one admissible outcome among many. We make no claim about human populations.

5. Conclusions

Simple interaction rules qualitatively reproduce much of what Calhoun observed in Universe 25. Crowding damage that gates breeding and care, and a long life span, give the collapse when adults made non-breeding by crowding do not recover. When adults recover on their own above a threshold rate, the colony resumes breeding and survives, with large, persistent, slowly damped

oscillations instead of collapsing; that threshold decreases with colony size. The collapse needs neither space nor the four social rules of the model; social learning in a juvenile window, retreat sites and crowd aversion add the failure of late-born cohorts, a persistent class of withdrawn non-breeders and the timing of the end of natality, and space adds empty nest sites at a low peak and a prolonged stagnation. Given a long life span, whether adults resume breeding once crowding is relieved, rather than density itself, is what separates the two outcomes, and it is what an experiment should measure.

CRediT authorship contribution statement

Julián Fernández Bonder: Conceptualization, Methodology, Software, Validation, Writing – original draft, Writing – review & editing, Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data and code availability

All code, simulation outputs and analysis scripts are available at `https://github.com/jfbonder/universe25-abm` and archived in Zenodo (Fernández Bonder, 2026). The model is the Python package `u25` (preset `calhoun_C1` for the reference model). The reduced time series behind Figs. 2, 4(a–d) and 6(a–d) (`results_em/series_deposit/`, 2.3 MB) and the per-run summaries of every ensemble are deposited with the code; the figure scripts regenerate every figure from them, and the ensemble runners regenerate the full series from the recorded seeds. The repository also documents the development, testing and analysis of the model (SM, Section S2).

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author used Claude (Anthropic), through Claude Code, in order to draft the text and to prepare the figures from the simulation outputs; the use of AI in the research itself is described in Section 2. The bibliographic data were checked against publisher and index records, and the reference list was reviewed by the author. No AI tool was used to create or alter images; the figures are plots of simulation data produced by the code. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

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